



Super Keyword in Java:

1. Super keyword is non static keyword in java.
2. The **Super keyword** in java is a reference variable which is used to refer immediate parent class object.

Uses of Java Super Keyword:

- Super can be used to refer immediate parent class instance variable.
- Super can be used to invoke immediate parent class method and variable as well.
- Super() can be used to invoke immediate parent class constructor.
- The most common use of the super keyword is to eliminate the confusion between super-classes and sub-classes that have methods with the same name.



CJC

Complete Java Classes

by Kunal Sir



**Super used to refer Immediate
parent class instance variable**

**super Used to invoke immediate
parent class method.**

**super() Used to invoke immediate
parent class constructor.**





Super Keyword in Case of Constructor Calling:

- It is used for calling parent class constructor.
- Every constructor has by default first line is super keyword.
- If we want to make call the parent class parameterized constructor then we want to write the explicit super keyword for calling that constructor.
- When want to call the constructor from super keyword, we must want follow the syntax like below-

super(); // For non-parameterized constructor

super(parameter); // For parameterized constructor

Understanding Program:

```
public class A
{
    public A ()
    {
        System.out.println("Constructor - A");
    }

    public A (String s)
```



CJC

Complete Java Classes

by Kunal Sir

```
{  
    System.out.println("Parameterized Constructor - A = "+s);  
}  
}
```

public class B extends A

```
{  
    public B ()  
    {  
        System.out.println("constructor - B");  
    }  
  
    public B (String s1)  
    {  
        super(s1);  
        System.out.println("constructor - B");  
    }  
}
```

public class Test

```
{  
    public static void main (String [] args)  
    {  
        B b = new B ();  
        B b1 = new B ("CJC");  
    }  
}
```



```
}  
}
```

Super Keyword in Case of Method Calling:

- It is used for calling parent class methods.
- We can call multiple methods from the super keyword.
- If we want to call the methods from super keyword, we must want follow the syntax like below-

super.methodName(); //For non-parameterized method

super.methodName(parameter); //For parameterized method

Understanding Program:

```
public class A  
{  
    public void m1()  
    {  
        System.out.println("M1- A");  
    }  
  
    public void m1(String s)  
    {  
        System.out.println("Parameterized M1- A = "+s);  
    }  
}
```



CJC

Complete Java Classes

by Kunal Sir

```
    }  
}  
  
public class B extends A  
{  
    public B ()  
    {  
        System.out.println("constructor - B");  
        super.m1();  
        super.m1("cjc");  
    }  
}  
  
public class Test  
{  
    public static void main (String [] args)  
    {  
        B b = new B ();  
    }  
}
```

Super Keyword in Case of Variable Calling:

- It is used for calling parent class variables.



CJC

Complete Java Classes

by Kunal Sir

- We can call multiple variables from the super keyword.
- If we want to call the variable from super keyword, we must want follow the syntax like below-

super.variableName;

Understanding Program:

```
public class A
{
    int x = 10;
    int y = 20;
    int z = 30;

    public A( )
    {
        System.out.println("Constructor- A");
    }
}
```

```
public class B extends A
{
    public B ( )
    {
```



CJC

Complete Java Classes

by Kunal Sir

```
System.out.println("constructor - B");  
System.out.println(super.x);  
System.out.println(super.y);  
System.out.println(super.z);  
}  
}
```

```
public class Test  
{  
    public static void main (String [] args)  
    {  
        B b = new B ();  
    }  
}
```

Note –

If we want to call the constructor by the help of super keyword, we can write the super keyword implicit / explicit first line of the constructor.

For the constructor call we can write only one super keyword inside the constructor.

For the methods and variable call, we can write multiple super keywords inside the constructor or method.